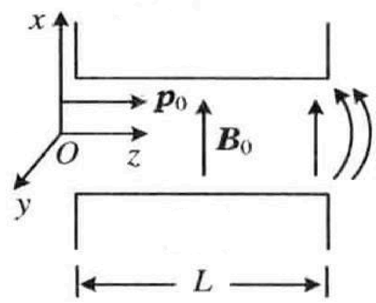
APhO 2005 T2B: Magnetic lens (improved)

This is adapted from a Chinese textbook (物理学难题集萃，舒幼生 et al.). The original APhO problem was worded really poorly and used some weird artificial assumptions that aren't really necessary or correct. Don't reference the original problem, as it will spoil some parts of the solution (even the title of the problem is a spoiler!). The difficulty of this problem is around a EuPhO T2.

Define a coordinate system as shown. We have two identical cuboidal permanent magnets with uniform magnetisation along the x-axis. The length of the magnets in the z-direction is L. The lengths of the magnets in the $\pm x$ direction and $\pm y$ direction are much larger than L. The magnets are placed with opposing poles close to each other (separation $\ll L$) to create a magnetic field with magnitude B_0 at the center of the cross-section.



A stream of charged particles, each with charge q, is ejected at $y = z = 0$ at various positions along the x-axis. The particles each have an initial momentum of $\boldsymbol{p_0} = p_0 \hat{\boldsymbol{z}}$, where $p_0 \gg qLB_0$. Show that the charged particles are focussed to a point, and find the z-coordinate of this point.

Ignore mutual interactions between the charged particles. The particles are moving at a nonrelativistic speed.

Solution

We can analyse the motion of the particles in two distinct phases.

1. The particles are deflected by the magnetic field $B_0\hat{\boldsymbol{x}}$, so its velocity has a component v_y in the y-axis when it reaches the edge of the magnets at $z = L$.
2. The fringe fields of the magnet close to $z = L$ provide a component of the magnetic field in the z-direction, which can interact with the y-component of the particle's momentum to deflect the particle in the x-direction, thus focussing the particles to a point.

It turns out that we can treat these two phases separately, as the fringe field is only significant close to $z = L$, and its extent is on the order of the separation between the magnets. Hence, we can ignore the fringe field when finding v_y, as the magnetic field is uniform within most of the region $z \in (0, L)$. Conversely, to analyse the focussing effect due to the fringe field, we can ignore any changes in v_y since this process only occurs very close to $z = L$.

Let us analyse this quantitatively. The cyclotron frequency is $\omega = qB_0/m$, so the angular deflection of the particle in the y-z plane along the y direction when the particle is at $z = L$ is

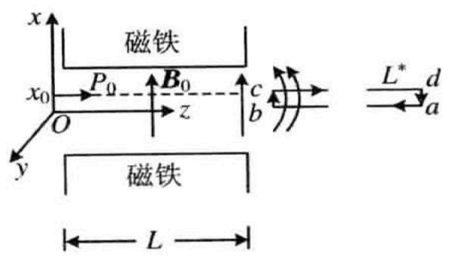
$$\theta_y = \frac{qB_0}{m}t \approx \frac{qB_0}{m}\frac{L}{v_0} = \frac{qB_0L}{p_0} \tag{1}$$

Using Newton's 2nd law and the small angle approximation,

$$m\frac{\mathrm{d}v_x}{\mathrm{d}t} = qv_yB_z \approx qv\theta_yB_z \tag{2}$$

$$\therefore v_x = \frac{qv\theta_y}{m}\int B_z\,\mathrm{d}t = \frac{q\theta_y}{m}\int B_z\,\mathrm{d}z \tag{3}$$

This integral can be found using Ampere's law. Create an Amperian loop as shown. Note that the diagram is a little misleading; cb should be in the gap between the magnets, at $z = L - D$, where $L \gg D \gg$ separation. Meanwhile, da is at $z = L + D$. Hence, the magnetic field is $B_0\hat{\boldsymbol{x}}$ at cb, while it is 0 at da. Thanks to Kevin Zhou for pointing this out!



For a particle with an initial x-coordinate of x_0,

$$\oint \boldsymbol{B}\cdot\mathrm{d}s = B_0x_0 + \int B_z\,\mathrm{d}z = 0 \tag{4}$$

$$\therefore \int B_z\,\mathrm{d}z = -B_0x_0 \quad\Rightarrow\quad v_x = -\frac{q\theta_y}{m}B_0x_0 \tag{5}$$

Using similar triangles,

$$\frac{x_0}{f} \approx \frac{x_0}{f-L} = -\frac{v_x}{v_0} \quad\Rightarrow\quad f = \frac{p_0^2}{q^2B_0^2L} \tag{6}$$